RTO Guard: what to do with risky cash-on-delivery orders

To: Founder | **From:** Ahad | **Date:** 1 November 2026

The short version. About 28% of our cash-on-delivery (COD) orders come back undelivered (RTO). Each one costs us ₹169: ₹65 to ship out, ₹65 to ship back, ₹25 handling, and ₹14 for the stock that comes back damaged. Across the 1,984 past orders I analysed, that is ₹95,711 lost. RTO Guard checks every new COD order and picks one of four actions: ship as-is, ship after WhatsApp confirmation, nudge to prepaid, or hold. Run on September (trained only on earlier months), it would have netted about **₹2,141** on 329 orders, roughly 15% of that month's ₹14,568 return loss, after counting the ad money lost on first-time customers. At your real volume (about 1,650 COD orders a month) that is roughly ₹10,000 a month. That scale-up is a rough guide, not a promise.

1. What surprised me in the data

- **Where the order is going matters most.** Patna, Darbhanga, Sagar, Gorakhpur and Bareilly return 48% to 62% of COD orders, against 18% in Mumbai and Pune. Those five towns are 14% of orders but about a quarter of all returns.
- **Fake details are a loud warning.** Orders with a junk address return 62% of the time, and orders with a junk name like "asdf" return 64%, against 25% and 28% for normal ones. Junk addresses are only 9% of orders but about one in five returns.
- **A customer's past is the best clue.** Customers with two earlier returns send back 76% of their orders. With one earlier return it is 35%, and with none it is 27%.
- **First-time customers are riskier, but not by much.** They return 30% against 23% for repeat customers, and they are 75% of all COD orders. A rule aimed at all new customers would block three orders in four to close a 7-point gap.
- **Some things barely matter.** Time of day (25% to 30%) and courier (26% to 31%) hardly move the return rate, so switching couriers will not fix this.

2. Recommended rules, and what they would have done to September

Each order gets a risk score, and the score decides the action. The cut-offs live in `config.yaml`, so Operations can change them without touching code.

- **Ship as-is:** low risk.
- **Nudge to prepaid:** medium risk. We offer about ₹30 off (4.6% of a ₹649 order). The maths: prepaid orders return only 4% of the time, so for an order with 30% risk, paying upfront saves us (30% − 4%) × ₹169 = ₹44 on average. We give back 70% of that and keep the rest as a buffer.
- **Confirm on WhatsApp first:** high risk. If the customer does not reply in 24 hours, the order is cancelled.
- **Hold or cancel:** only the very highest risk.

What this did to September's 329 orders:

Action	Orders	Actually returned	Net ₹
Ship as-is	94	9%	0
Nudge to prepaid	210	29%	+1,372
Confirm first	19	63%	+686
Hold / cancel	6	83%	+83
Total	**329**		**+2,141**

Total saved was ₹4,646 and total lost was ₹2,505. Most of the money comes from nudging and confirming, not from cancelling: the six holds earned only ₹83.

3. Operations vs Marketing, in rupees

Operations wants every risky order cancelled. Rough maths, using your ₹649 average order and 62% margin (about ₹402 profit per delivered order): cancelling all 235 flagged September orders would avoid about 78 returns (about ₹13,200) but also throw away about 157 genuine orders (about ₹63,000 of profit). That is roughly ₹50,000 worse than doing nothing, because most "risky" orders still arrive: 37% of the confirm group and 71% of the nudge group.

Marketing wants first orders protected, and the numbers agree. Of 241 first-time customers in September, only 6 (2.5%) were held. 56 shipped untouched, 15 were asked to confirm, and 164 were offered a prepaid discount. Every time a genuine ad-sourced first-

time customer is lost, we count the ₹300 ad cost on top of the lost profit (a smaller share when they only drop out at confirmation). That added up to ₹450 in September, and the net was still ₹2,141.

The trade-off. The six holds earned only ₹83. If Marketing wants a rule of "never cancel a first order", it would cost about ₹83 on September's volume. That is a cheap price for the protection, and I would offer it.

4. What could go wrong, and how we would notice

- **The ₹ figures rest on two assumptions:** that confirmation stops 60% of returns and that 30% of customers accept the prepaid offer. If the real rates are lower, the net shrinks. We would notice by recording every WhatsApp reply and every accepted offer from day one.
- **One month is thin evidence, and the rules were tuned on it.** Each week, check that "ship as-is" orders still return at about 9%. If that climbs above roughly 15%, the model has gone stale.
- **Discounts can leak.** 71% of nudged orders would have arrived anyway, and the offers cost ₹1,208 in September. Customers may also learn to wait for a discount. Watch the share of orders getting a discount and the weekly discount total.
- **The small buckets are noisy.** With 6 holds and 19 confirmations, one order moves the rate by 5 to 17 points. Do not judge by a single week.
- **Known blind spots:** near-duplicate orders are not detected, the AI address check has not been tested on the live service, and fake-name rules are easy to dodge. A rise in returns among "ship as-is" orders would be the signal.
- **Silence cancels confirmation orders.** Some genuine customers will simply miss the message. Letting a small test group ship anyway would show how many.

5. What I would build next (two more weeks)

1. A live pilot with a holdout group, to measure the real confirmation and prepaid acceptance rates instead of assuming them.
2. Monthly rolling backtests, plus a table showing how the net changes if the assumptions are off.
3. Detection of near-duplicate orders placed minutes apart.
4. A test of the AI address check on the live service against a hand-labelled sample.

5. A connection to Shopify and the real WhatsApp Business service, so decisions happen automatically when an order arrives.